

Thermal and Temporal Effects on UHT Milk During Storage

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INTRODUCTION

Flavour is strongly influenced by aroma, making the analysis of volatile compounds an important aspect of understanding food flavour. Fresh milk is typically characterised by mild, fresh and fatty aromas. However, the high thermal load during ultra-high temperature (UHT) processing can promote Maillard reactions that create cooked, roasted and burnt aromas.

While Maillard products form during UHT processing, information on how post-process handling influences the final product remains limited. During retail storage, air-conditioned environments may maintain milk at approximately 20 °C. In contrast, during transport, especially export or regional distribution, UHT milk may be exposed to substantially higher temperatures for extended periods. The extent to which these different storage temperatures alter the volatile composition, and potentially the flavour, of UHT milk remains unclear. This study investigates ongoing volatile changes in UHT milk stored at different temperatures for 1 month, using comprehensive multidimensional gas chromatography (GC×GC), which provides a more complete profile of milk volatiles than has previously been reported, to identify changes that may affect flavour.

METHODS

Full-cream UHT milk was stored at 20 °C and 40 °C for 1 month directly after UHT processing. Samples were collected weekly for comprehensive volatile analysis using a LECO HS-SPME—GC×GC—ToFMS. Comparison of the profiles was conducted using GC×GC alignment software to perform class analysis and principal component analysis (PCA).

RESULTS AND DISCUSSION

PCA revealed distinct clusters for nearly all sample types, with some crossover between clusters for milks stored for one week at 20 °C and 40 °C, respectively. Following the PCA, individual features were compared, and it was found that several Maillard products showed different trends with time and temperature.

One example is 2-ethylfuran, a Maillard product from the furan family, which remains stable in abundance during 20 °C storage but increases at 40 °C, as illustrated in Figure 1. This is notable because 2-ethylfuran has coffee- and malty aromas at low

abundances but rapidly becomes harsh and chemical at higher concentrations [1, 2]. As this, and other Maillard products, increase in concentration, the overall aroma and flavour of milk may be significantly altered.

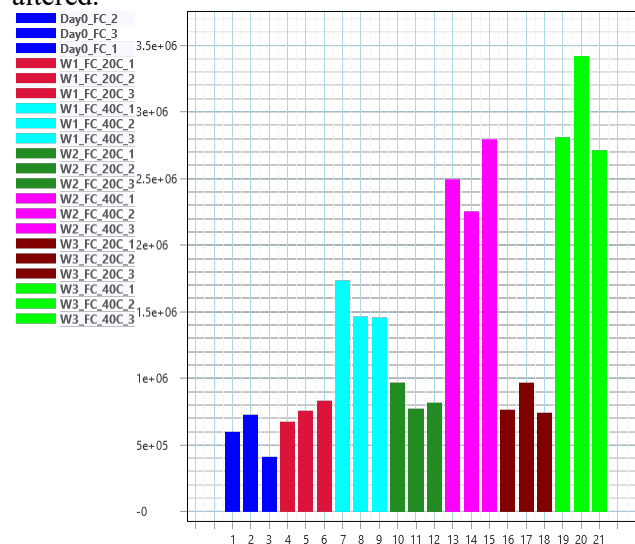


Figure 1. 2-Ethylfuran abundance during storage at different temperatures.

CONCLUSIONS

Both storage temperature and duration can significantly affect the volatile profile of UHT milk by promoting Maillard reactions initiated during thermal processing. The resulting Maillard products can affect final product flavour, potentially reducing consumer acceptance and contributing to food waste.

ACKNOWLEDGEMENTS

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REFERENCES

- [1] The Good Scents Company [2-ethyl furan, 3208-16-0](#)
- [2] [FlavorDB](#)